Past and Future of Cosmic Velocities

Arthur Kosowsky
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… and grad school office-mate of Andrew Jaffe, 1991-1993

Linear perturbation theory:

$$\frac{\partial \delta}{\partial t} + \frac{1}{a} \nabla \cdot \mathbf{v} = 0$$

$$\mathbf{k} \cdot \mathbf{v} = \left[iaHf(\Omega_m) \right] \delta$$

THE ASTROPHYSICAL JOURNAL, 515:L1–L4, 1999 April 10

STREAMING VELOCITIES AS A DYNAMICAL ESTIMATOR OF Ω

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Received 1998 August 8; accepted 1999 February 5; published 1999 March 3

ABSTRACT

It is well known that estimating the mean pairwise velocity of galaxies v_{12} from the redshift space galaxy correlation function is difficult because this method is highly sensitive to the assumed model of the pairwise velocity dispersion. Here we propose an alternative method to estimate v_{12} directly from peculiar velocity samples, which contain redshift-independent distances as well as galaxy redshifts. In contrast to other dynamical measures which determine $\beta \equiv \Omega^{0.6}\sigma_8$, this method can provide an estimate of $\Omega^{0.6}\sigma_8^2$ for a range of σ_8, where Ω is the cosmological density parameter, while σ_8 is the standard normalization for the power spectrum of density fluctuations. We demonstrate how to measure this quantity from realistic catalogs.

Subject headings: cosmology: observations — cosmology: theory — galaxies: distances and redshifts — galaxies: kinematics and dynamics

Evidence for a Low-Density Universe from the Relative Velocities of Galaxies

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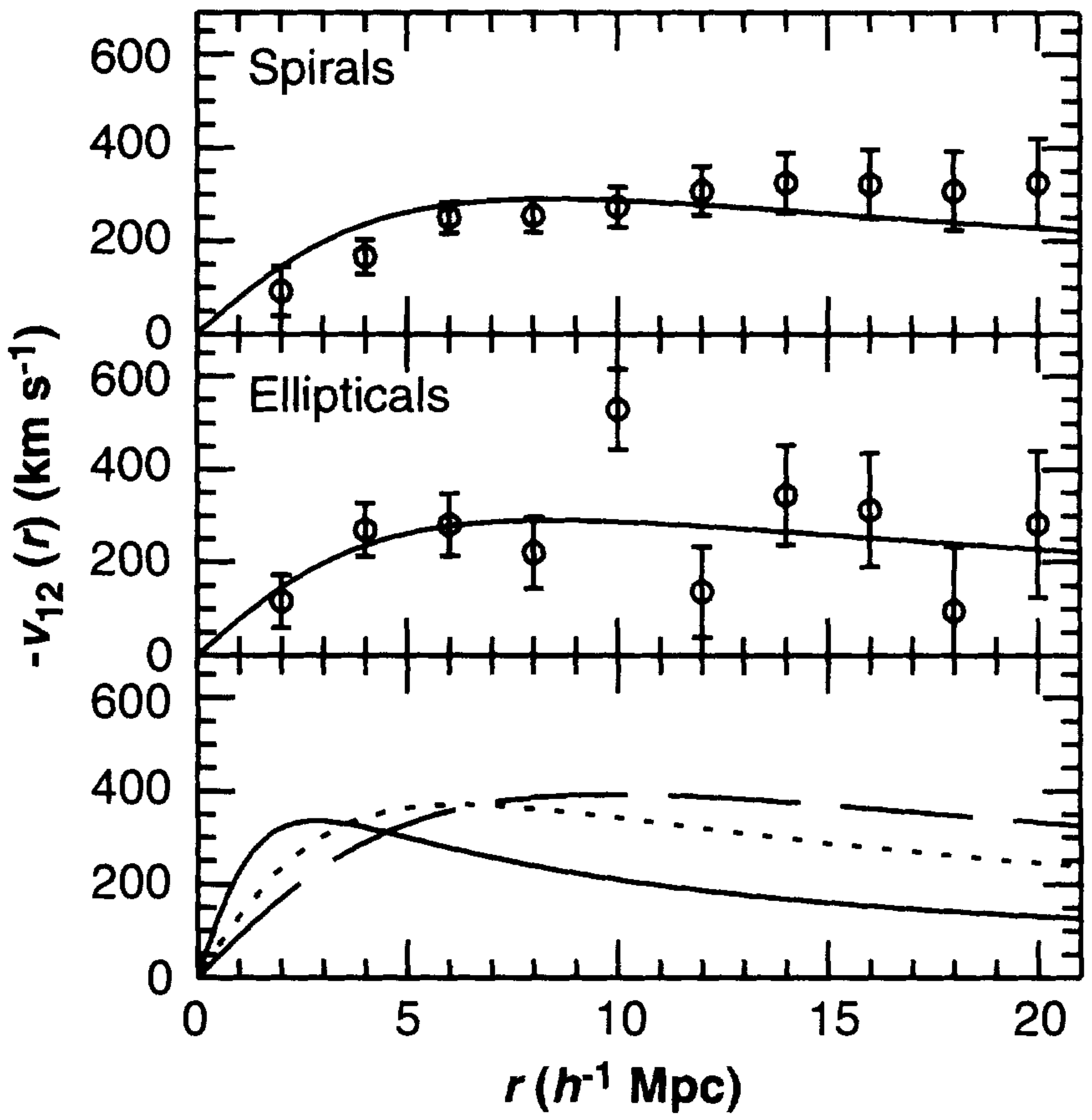
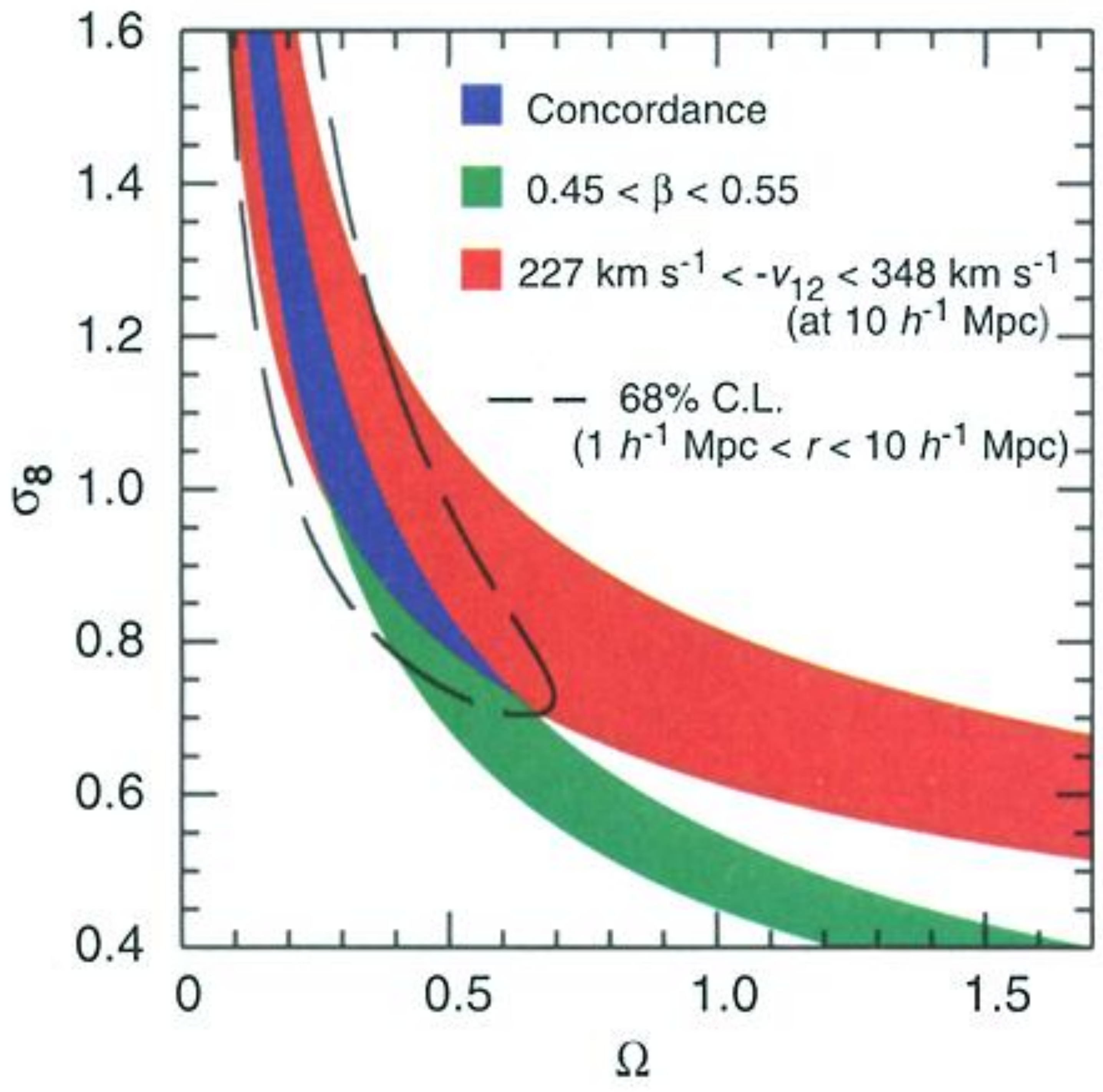
The motions of galaxies can be used to constrain the cosmological density parameter Ω and the clustering amplitude of matter on large scales. The mean relative velocity of galaxy pairs, estimated from the Mark III survey, indicates that $\Omega = 0.35^{+0.35}_{-0.25}$. If the clustering of galaxies is unbiased on large scales, $\Omega = 0.35 \pm 0.15$, so that an unbiased Einstein–de Sitter model ($\Omega = 1$) is inconsistent with the data.

The statistic we consider was introduced in the context of the Bogoliubov-Born-Green-Kirkwood-Yvon (BBGKY) kinetic theory describing the dynamical evolution of a self-gravitating collection of particles (*3, 6*). One of the BBGKY equations is the pair conservation equation, relating the time evolution of v_{12} to $\xi(r)$, the two-point correlation function of spatial fluctuations in the fractional matter density contrast (*3*). Its solution is well approximated by (*4*)

$$v_{12}(r) = -\frac{2}{3} Hr\Omega^{0.6}\bar{\bar{\xi}}(r)[1 + \alpha\bar{\bar{\xi}}(r)] \quad (1)$$

$$\bar{\bar{\xi}}(r) = \frac{3\int_0^r \xi(x)x^2dx}{r^3[1 + \xi(r)]} \quad (2)$$





The First Commandment of Modern Cosmology

Thou Shalt Use the Cosmic Microwave Background

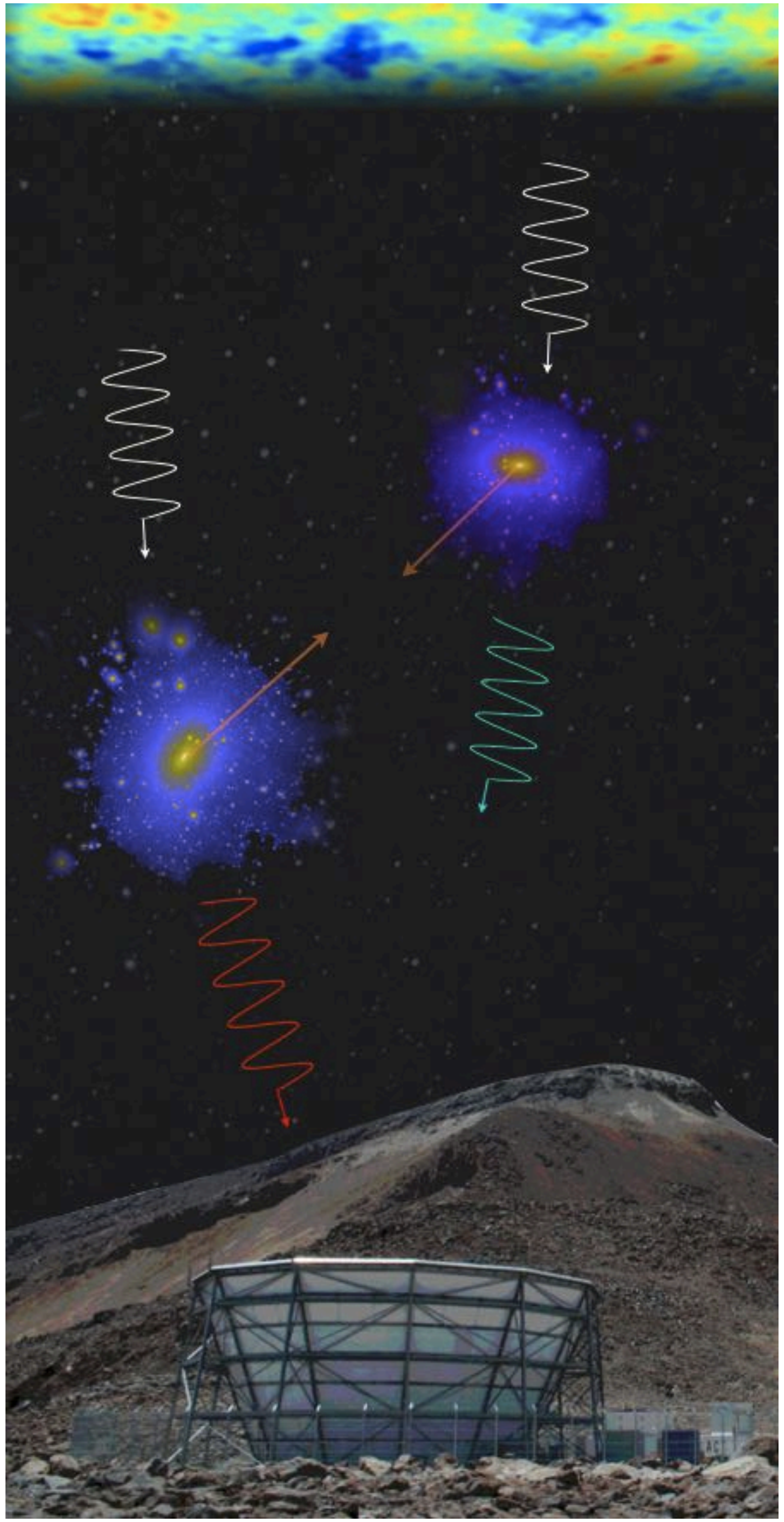
The kinematic Sunyaev-Zeldovich Effect

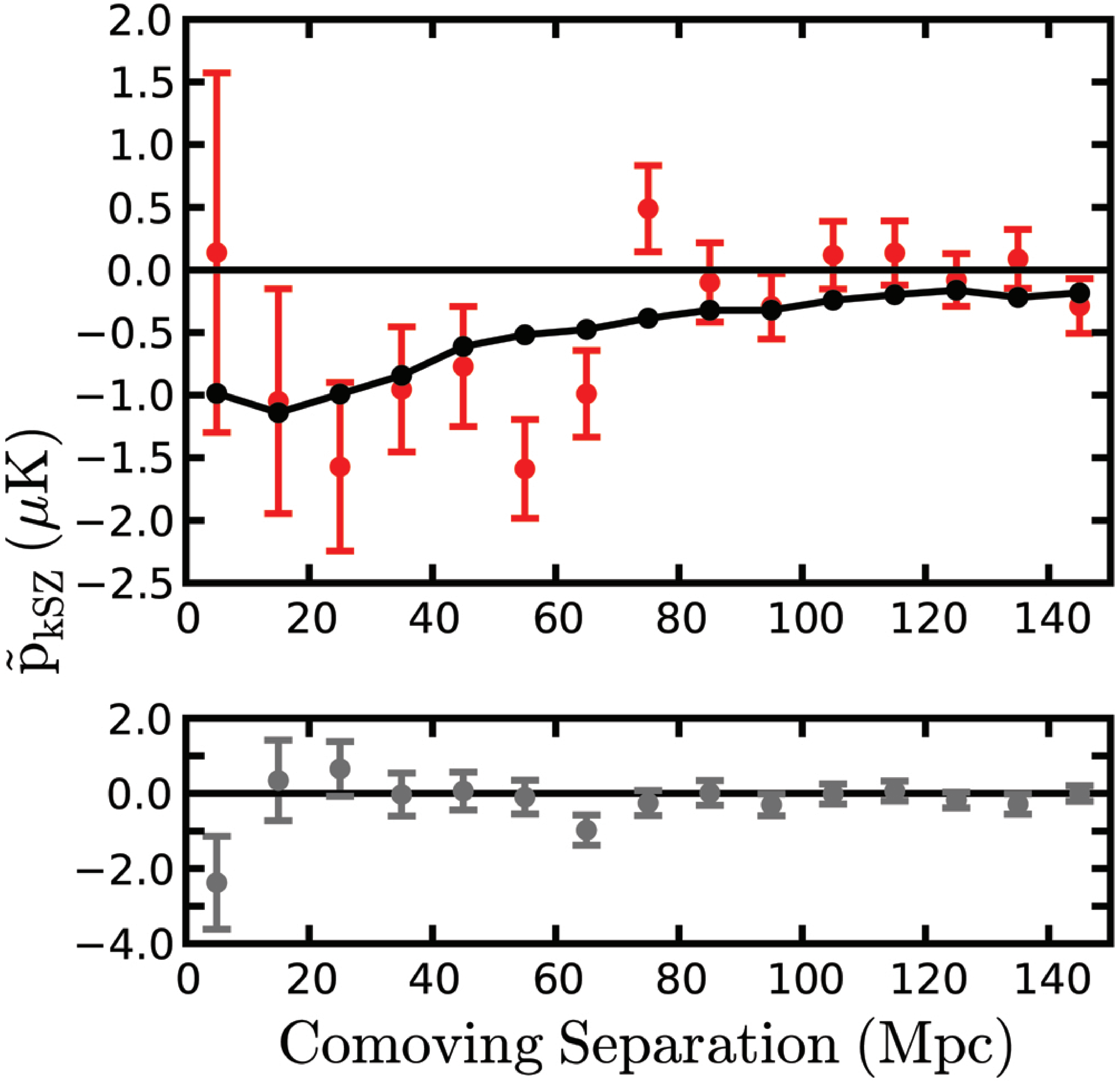
$$\frac{\Delta T}{T_0} \propto \int d\chi \, n_e v_{\rm los}$$

Line of sight integral of the ionized gas momentum

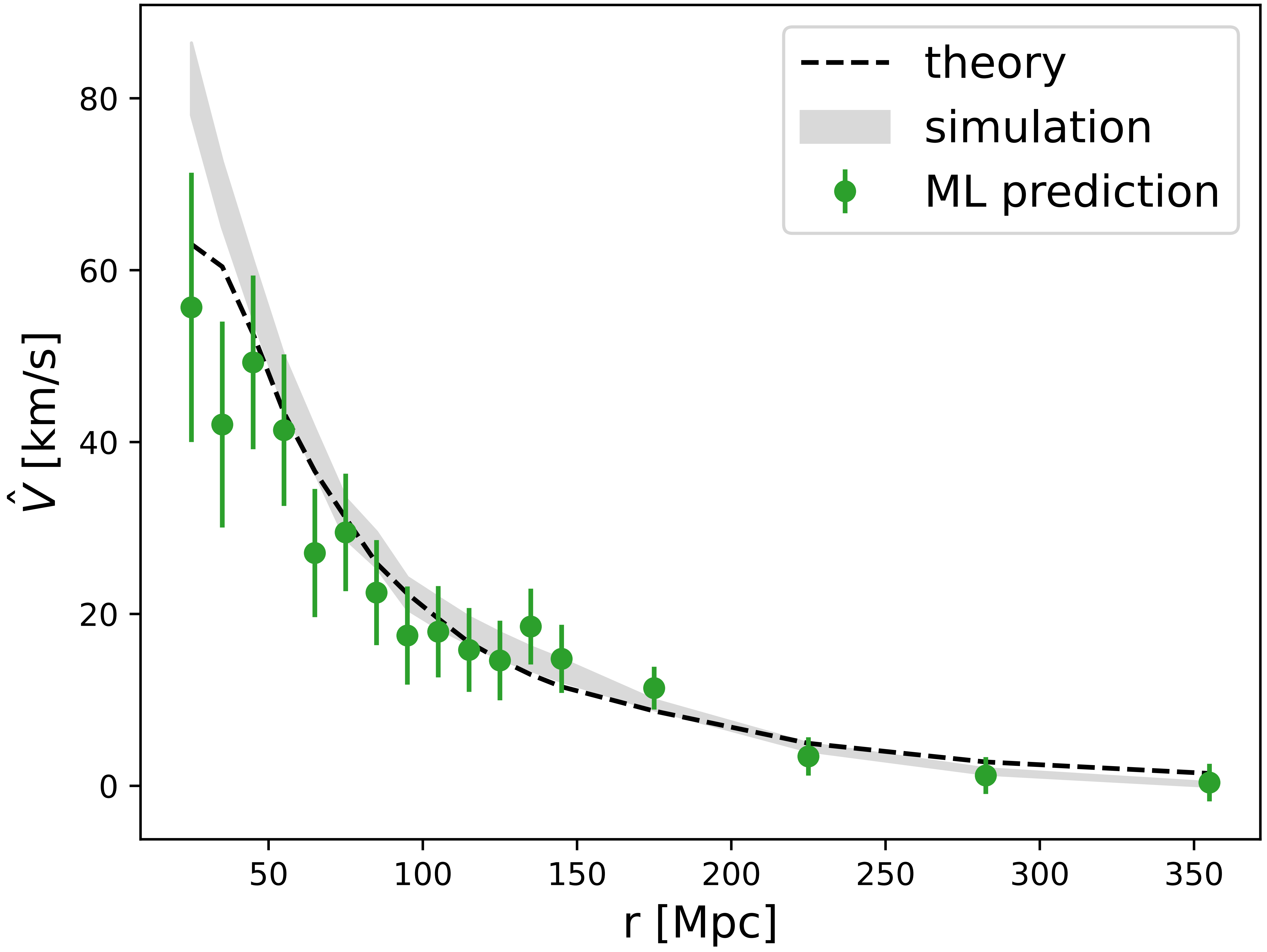
R. Sunyaev and Ya. B. Zel’dovich, Comments Astrophys. Space Physics 4, 173 (1972)

Small: few muK for 5e14 M_sun cluster, v = 300 km/s





N. Hand et al. PRL 109, 1101 (2012)
The first direct detection of cosmological peculiar velocities

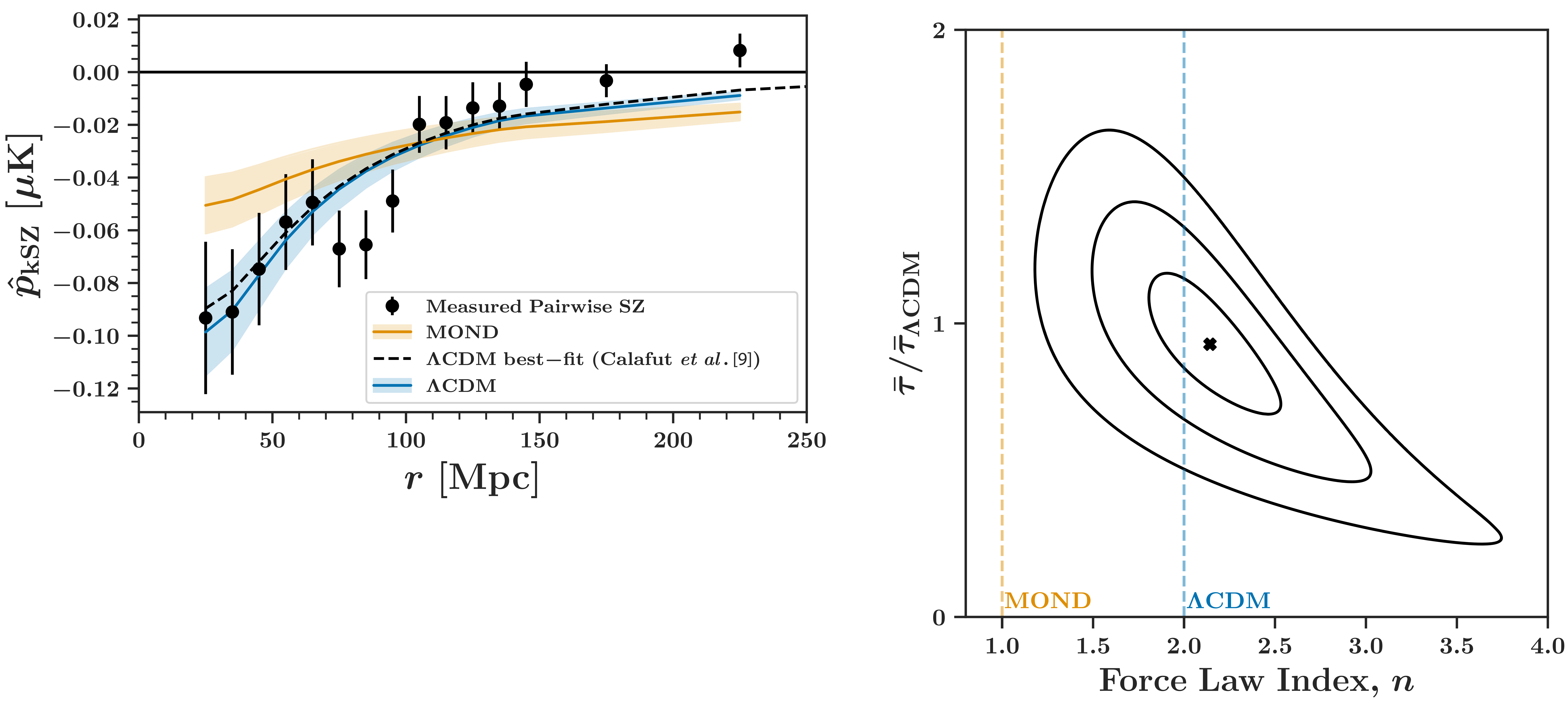


ACT DR6 + Planck
DESI DR1

913,000 Luminous Red Galaxies
457,000 clusters with M > 1e13 M_sun

9.3 sigma detection significance

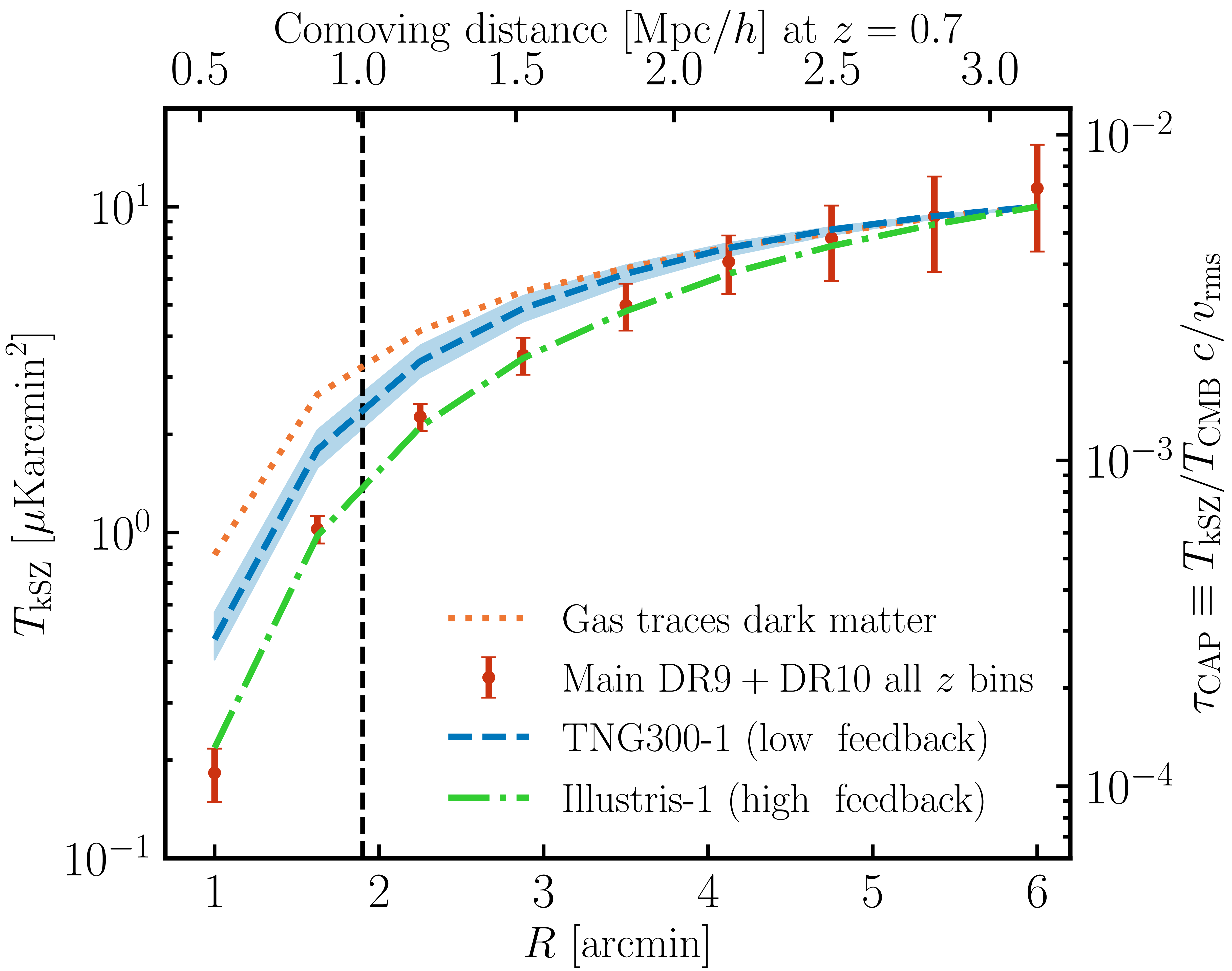
Y. Gong et al. PRD 113, 063538 (2026)



P. Gallardo, K. Pardo et al., PRL 136, 151002 (2026)

Galaxy density map: estimate 3d velocity field

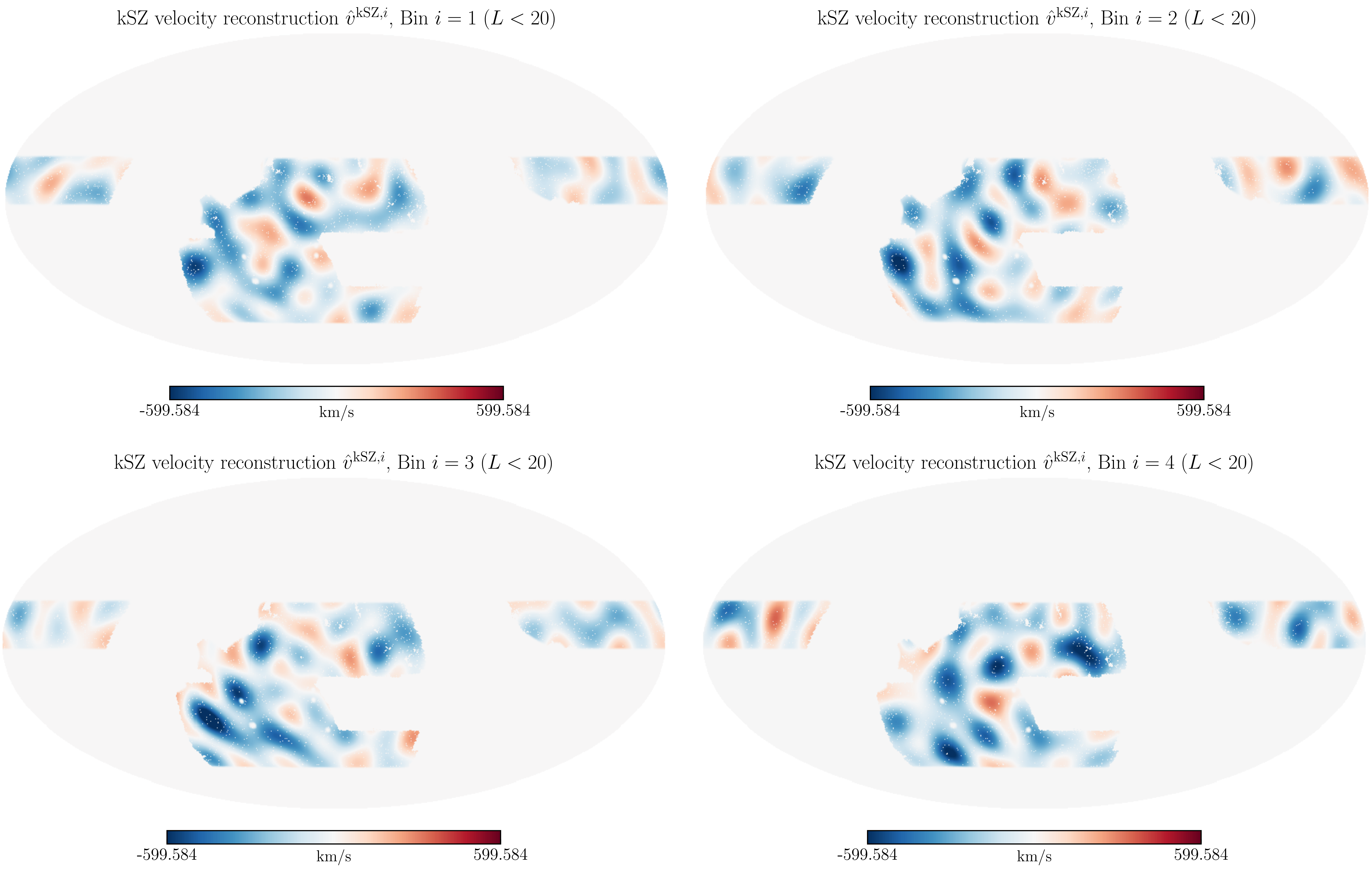
Stack CMB at positions of galaxies, weighted by estimated line-of-sight velocity



Hadzhiyska et al. PRD 112, 3509 (2025): ACT DR4 + DESI LRG's

Kinematic SZ Tomography

Estimate density and velocity from galaxy survey, correlate with CMB temperature



$$\langle ggT \rangle$$

ACT DR6
DESI LRG’s (photo-z)
BOSS (spec-z)

3.8 sigma significance

McCarthy et al.,
JCAP 2025, 05, 057

Moving Lens Effect

M. Birkinshaw and S. Gull, Nature 302, 315 (1983)

$$\frac{\delta T(\boldsymbol{n})}{T} = -\frac{2}{c^2}\int_{\chi_0}^{\chi_*} \boldsymbol{\nabla}\Phi(\boldsymbol{n})\cdot\frac{\boldsymbol{v}(\boldsymbol{n})}{c}d\chi \equiv \boldsymbol{\beta}(\boldsymbol{n})\cdot\frac{\boldsymbol{v}(\boldsymbol{n})}{c}$$

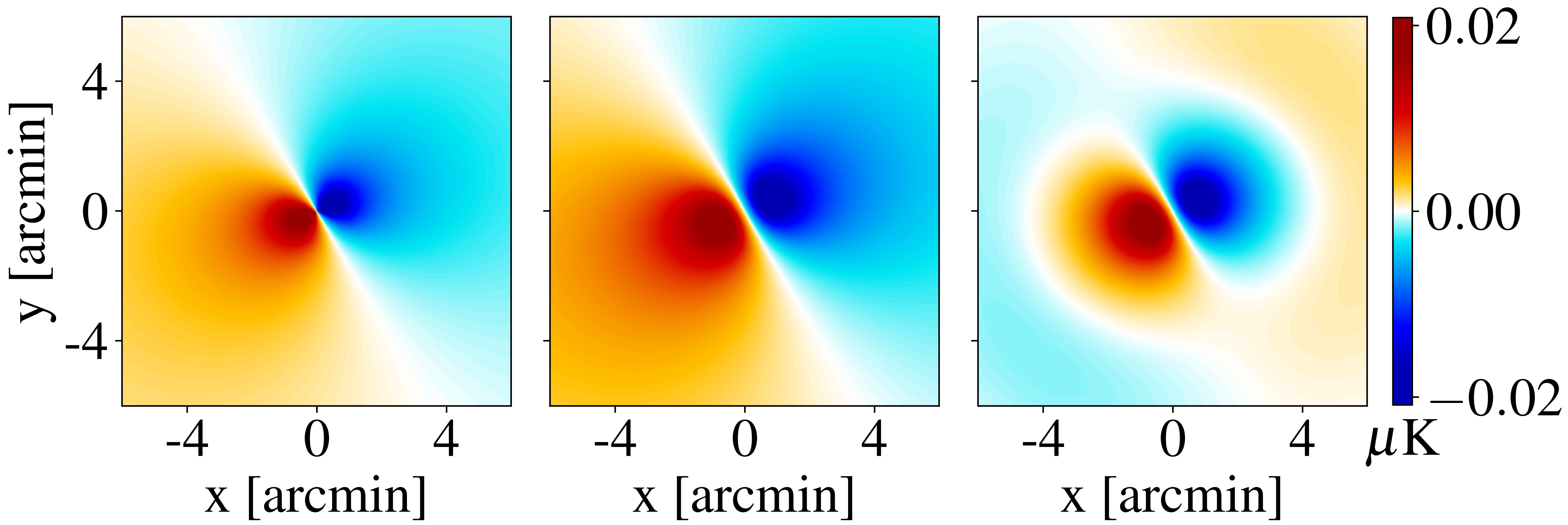
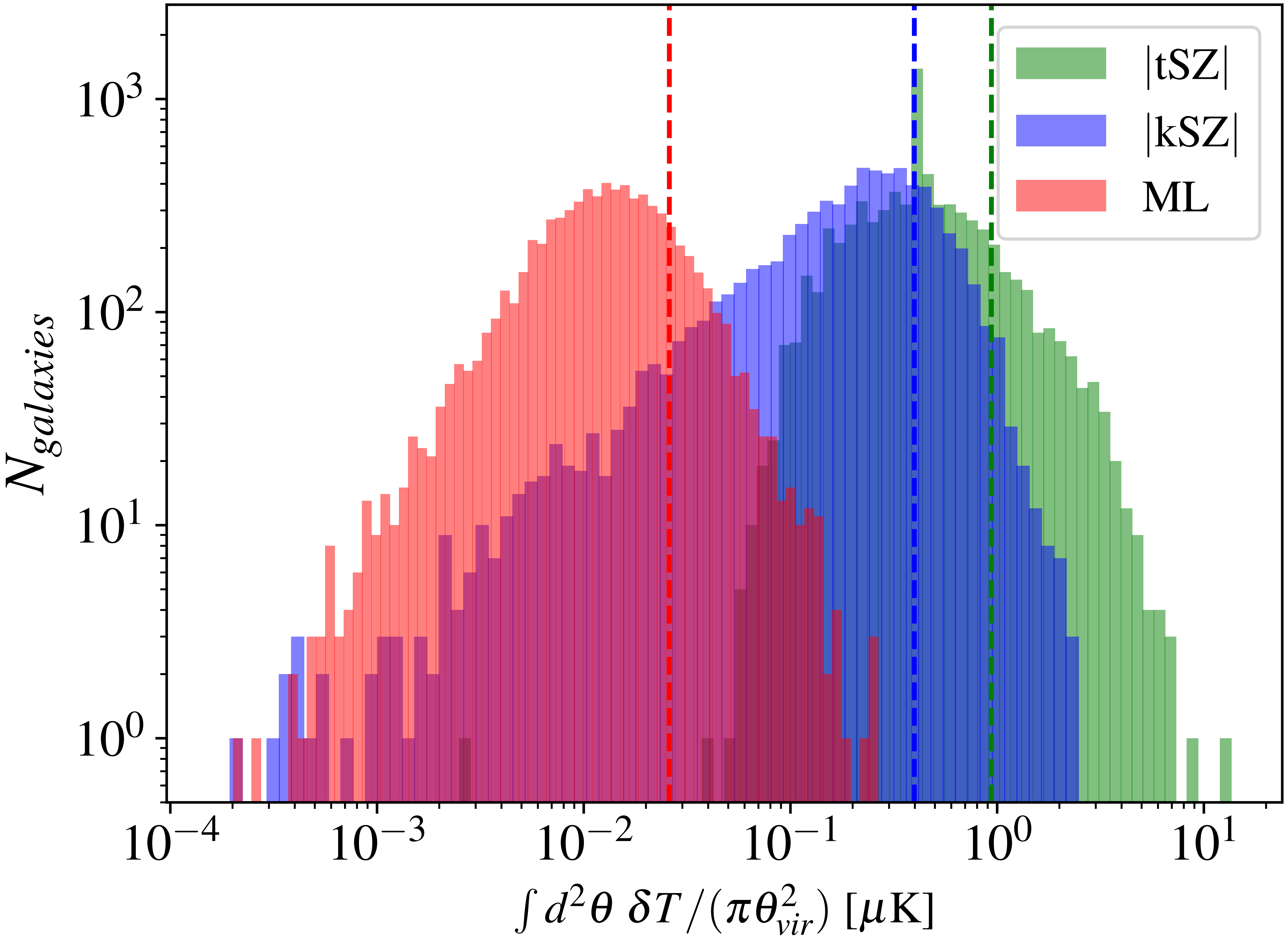


FIG. 4. Dipolar pattern of anisotropy caused by a moving cluster (left), 1' beamed (middle), 1' beamed and high-pass filtered with $\ell_{\rm cut} = 2000$ (right). This cluster has a mass of 5×10^{13} $M_\odot$, transverse velocity of 450 km/s and redshift $z = 0.5$.



Signal comparison for 100 square degrees of BOSS

A. Beheshti, E. Schaan, and A. Kosowsky, PRD 111, 3510 (2025)

Moving Lens Effect S/N

SNR	#Galaxies overlapping	ACT	SO	S4
BOSS	350k	1.2 (0.2)	1.4 (0.7, 1.4, 0.3)	1.8
DESI LRGs Y1	900k	1.8 (0.4)	2.3 (1.2, 2.2, 0.5)	2.8
DESI LRGs Y5	3M	3.4 (0.7)	4.3 (2.2, 4, 0.9)	5.1
DESI LRGs + Ext. LRGs Y1	3.4M	3.7 (0.8)	4.6 (2.3, 3.4, 1)	5.4
DESI LRGs + Ext. LRGs Y5	11.4M	6.4 (1.4)	8.3 (4.2, 7.8, 1.8)	9.9
unWISE	50.8M	11.9 (2.6)	14.8 (7.6, 14.0, 3.1)	17.8
LSST equal weight (pessimistic)	1.12B	10.2 (2.2)	12.7 (6.5, 11.9, 2.7)	15.1
LSST equal weight; $10^{12}M_\odot < M$	330M	18.0 (4.0)	22.5 (11.4, 21.1, 4.7)	26.8
LSST equal weight, $10^{14}M_\odot < M$	8.4M	51.6 (11.4)	64.6 (32.8, 60.4, 13.4)	76.7
LSST mass weight (optimistic)	1.12B	68.1 (15.1)	85.3 (43.2, 79.8, 17.6)	101.3

A. Beheshti, E. Schaan, and A. Kosowsky, PRD 111, 3510 (2025)

Simons Observatory + Rubin Observatory

Almost completely overlapping survey regions

Uniform map depths

Huge number of tracers, different galaxy subpopulations

Well-studied systematics

Coming Up: characterizing the 3-dimensional cosmic velocity field using kSZ (line of sight component) and moving lens (transverse components) at high statistical significance

Come to Andrew's 70th Birthday Conference